

Activity

Case Analysis – Avoiding Misleading Proportions

Question 1: Truncated Y-Axis in Sales Growth

A retail company presents the following monthly sales (in ₹ lakhs):

Month	Sales
January	95
February	97
March	99
April	101
May	103
June	105

Tasks:

1. Explain why the chart is misleading.

The chart is misleading because the Y-axis begins at 90 instead of 0, which exaggerates the visual difference between the sales values. The sales increase from ₹95 lakhs in January to ₹105 lakhs in June, representing an actual increase of only ₹10 lakhs (approximately 10.5%). However, because the axis is truncated, the June bar appears much taller than the January bar, creating the false impression that sales have nearly doubled when the increase is actually modest.

2. Identify the visualization principle that has been violated.

The visualization principle that has been violated is the zero-baseline principle for bar charts. Since the length of a bar represents the magnitude of a value, bar charts should generally start the Y-axis at zero. Beginning the axis at 90 distorts the comparison between the bars and misrepresents the true difference in sales.

3. Recommend an improved visualization method.

A more appropriate visualization would be a bar chart with the Y-axis starting at zero, as this would accurately represent the differences in sales across the months. Alternatively, a **line** chart could be used to show the steady upward trend in monthly sales, since line charts are better suited for displaying changes over time without exaggerating the magnitude of the increase.

4. Describe how this misleading chart could influence business decisions.

This misleading chart could cause managers and stakeholders to believe that sales growth is much greater than it actually is. As a result, they might overestimate the effectiveness of marketing efforts, allocate larger budgets based on inaccurate conclusions, set unrealistic future sales targets, or present an overly optimistic view of the company's performance to investors and other stakeholders. Accurate visualizations are essential because they support informed and reliable business decision-making.

Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment in four departments.

Department	Students
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Computer Science	420
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Electronics	280
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Mechanical	200
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Civil	100
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Tasks:

1. Calculate the correct percentage for each department.

The total number of students is 1000 (420 + 280 + 200 + 100). Therefore, the correct percentage for each department is calculated by dividing the number of students in each department by the total and multiplying by 100. Computer Science has 420 students, which represents 42% of the total enrolment. Electronics has 280 students, accounting for **28%**. Mechanical has 200 students, making up 20%, while Civil has 100 students, representing 10% of the total.

2. Explain why the displayed chart is misleading.

The displayed pie chart is misleading because the slice sizes do not match the actual proportions of student enrolment. The chart incorrectly shows the departments as 50%, 30%, 15%, and 15% instead of the correct values of 42%, 28%, 20%, and 10%. This exaggerates the share of Computer Science and Civil while understating the share of Mechanical. As a result, viewers receive an inaccurate understanding of how students are distributed across the departments.

3. Suggest an alternative chart that would improve comparison accuracy.

A bar chart would be a better alternative because it allows viewers to compare the number of students across departments more accurately using the lengths of the bars. If a pie chart is still preferred, it should be drawn using the correct percentages of 42%, 28%, 20%, and 10% so that each slice accurately represents its share of the total enrolment.

4. Discuss how such errors could affect institutional planning.

Errors in the chart could lead university administrators to make poor planning decisions. They might allocate faculty, classrooms, laboratory facilities, funding, or other resources based on incorrect enrolment proportions. Departments that appear larger than they actually are may receive unnecessary resources, while departments that are underrepresented in the chart may not receive adequate support. Such inaccurate visualizations can also mislead prospective students, faculty members, and stakeholders, resulting in ineffective planning and resource allocation.

Question 3: Unequal Bar Widths in Revenue Comparison

A software company compares annual revenue.

Product Revenue (₹ Crores)

A	60
B	75
C	90
D	105

Tasks:

1. Identify the graphical error.

The graphical error is the use of unequal bar widths in the bar chart. In a standard bar chart, all bars should have the same width so that only the height represents the data values. By making the bars progressively wider, the chart exaggerates the visual importance of certain products. This causes Product D to appear much larger than Product A, even though the actual revenue difference is relatively modest.

2. Explain how unequal bar widths distort perception.

Unequal bar widths distort perception because viewers tend to judge the overall size of a bar by its **area**, which is influenced by both height and width. When the width of a bar is increased

along with its height, the bar appears disproportionately larger. In this case, Product D has a revenue of ₹105 crores compared to ₹60 crores for Product A, which is only 1.75 times greater. However, the increased width makes Product D appear nearly three times larger, giving a false impression of the actual revenue difference.

3. Redesign the visualization by specifying the correct design principles.

The chart should be redesigned using equal-width bars for all products, with only the heights representing the revenue values of ₹60 crores, ₹75 crores, ₹90 crores, and ₹105 crores. The Y-axis should begin at zero and use a consistent scale with clearly labelled values. Proper spacing between the bars, accurate axis labels, and a descriptive title should also be included to ensure that the visualization presents the revenue data clearly and without distortion.

4. Discuss the possible consequences if investors rely on this chart.

If investors rely on this misleading chart, they may believe that Product D generates significantly more revenue than it actually does compared to Product A. This could lead to incorrect investment decisions, unrealistic expectations about the company's product performance, and poor allocation of financial resources. The misleading visualization could also reduce trust in the company's reporting if the distortion is discovered, potentially affecting investor confidence and the organization's credibility.

Question 4: 3D Column Chart Distortion

An energy company reports renewable energy generation.

Source Energy (GWh)

Solar 450

Wind 500

Hydro 550

Biomass 520

Tasks:

1. Explain why 3D charts can create misleading proportions.

3D charts can create misleading proportions because perspective and viewing angles make some bars appear larger or smaller than they actually are. The depth effect can exaggerate differences between values, making it difficult for viewers to accurately compare the heights of the columns.

2. Compare the perceived difference with the actual difference.

The chart gives the impression that Hydro (550 GWh) is almost 40% larger than Wind (500 GWh) because of the 3D perspective. In reality, the difference is only 50 GWh, which represents an actual increase of 10%, showing that the visual exaggeration is much greater than the true data difference.

3. Recommend a better visualization.

A simple 2D column chart or bar chart would be a better choice because it represents the values without perspective distortion. This allows viewers to compare the data accurately and understand the actual differences between the energy sources.

4. Discuss the importance of avoiding visual distortion in sustainability reporting.

Avoiding visual distortion in sustainability reporting is essential because stakeholders rely on charts to make informed decisions. Misleading graphics can exaggerate or minimize performance, reducing the credibility of the report and potentially leading to incorrect conclusions about renewable energy generation. Clear and accurate visualizations help maintain transparency, trust, and effective communication of sustainability data.

Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization compares disease cases across five districts.

District Cases

A	150
B	250
C	350
D	450
E	550

Tasks:

1. Explain why scaling the radius instead of the area is misleading.

Scaling the bubble radius directly to the number of cases is misleading because the area of a circle increases with the square of its radius. This causes larger values to appear much bigger than they actually are, exaggerating the differences between districts. As a result, District E looks more than ten times larger than District A even though it has only about 3.7 times as many cases (550 compared to 150).

2. Describe the correct method for proportional bubble sizing.

The correct method is to scale the **area** of each bubble in proportion to the data values rather than the radius. This means the radius should be calculated using the square root of the value so that the bubble's area accurately represents the number of cases and allows fair visual comparison.

3. Suggest an alternative visualization suitable for this dataset.

A 2D bar chart or column chart would be a better visualization because it allows viewers to compare the number of cases across districts accurately without the risk of distortion caused by incorrectly sized bubbles.

4. Discuss the potential impact of this visualization on public health resource allocation.

An incorrectly scaled bubble chart can mislead decision-makers by making differences between districts appear much larger than they really are. This may result in resources such as medical staff, funding, or equipment being allocated unfairly, reducing the effectiveness of public health planning and potentially affecting patient care.